



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS

(Second paper)

NQF LEVEL 2

12 November 2025

MARKING LEGEND	
SYMBOL	EXPLANATION OF SYMBOL
(A)	Accuracy
M	Method
(CA)	Consistent accuracy (follow-up)
B/D	Breakdown
S	Statement
R	Reason
S/R	Statement and reason
F	Choosing correct formula
SF	Substitution into correct formula

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Half mark ✓ Full mark ✓

QUESTION 1

1.1						Explanation	Mark																				
	1.1.1		<table><thead><tr><th>Car brand</th><th>Tally</th><th>Frequency</th></tr></thead><tbody><tr><td>Toyota</td><td>HHH HHH HHH</td><td>15✓</td></tr><tr><td>VW</td><td>HHH</td><td>5✓</td></tr><tr><td>Hyundai</td><td>IIII</td><td>4✓</td></tr><tr><td>Mercedes</td><td>HHH</td><td>8✓</td></tr><tr><td>Ford</td><td>HHH</td><td>8✓</td></tr><tr><td></td><td>Total</td><td>40</td></tr></tbody></table>	Car brand	Tally	Frequency	Toyota	HHH HHH HHH	15✓	VW	HHH	5✓	Hyundai	IIII	4✓	Mercedes	HHH	8✓	Ford	HHH	8✓		Total	40		1 mark for each row with correct information	(5)
Car brand	Tally	Frequency																									
Toyota	HHH HHH HHH	15✓																									
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Mercedes	HHH	8✓																									
Ford	HHH	8✓																									
	Total	40																									
	1.1.2	Toyota ✓				1A	(1)																				
	1.1.3	A bar chart of car brands Frequency Car brand Toyota VW Hyundai Mercedes Ford				1 mark for each bar with the correct height	(4)																				

1.2				Explanation	Mark
	1.2.1	13; 23; 37; 42; 53; 63; 71; 72; 72; 76; 84 ✓		1 mark for order	(1)
	1.2.2	$Q_2 = 63$	$Q_2 \text{ Position} = \frac{2}{4}(11 + 1) = 6^{th}$ $\therefore Q_2 = 63 \checkmark$	1CA	(1)
	1.2.3	$Q_1 \text{ Position} = \frac{1}{4}(11 + 1) = 3^{rd} \checkmark$ $\therefore Q_1 = 37 \checkmark$		1SF 1A	(2)
	1.2.4	$Q_3 \text{ Position} = \frac{3}{4}(11 + 1) = 9^{th} \checkmark$ $\therefore Q_3 = 72 \checkmark$		1SF 1A	(2)
	1.2.5	Range = $84 - 13 = 71 \checkmark$		1CA	(1)
	1.2.6	IQR = $72 - 37 = 35 \checkmark$		1CA	(1)

1.2.7	$P_{85} \text{ Position} = \frac{85}{100}(11 + 1) = 10,2^{th} \checkmark$ $\therefore P_{85} = 76 + 0,2(84 - 76) \checkmark$ $= 77,6 \checkmark$	1SF 1M 1A	(3)
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1.3			Explanation	Mark
1.3.1	Women = $\frac{21\ 000}{90\ 000} \times 360^\circ \checkmark$ $= 84^\circ \checkmark$		1M 1A	(2)
1.3.2	Children = $\frac{4\ 500}{90\ 000} \times 360^\circ \checkmark$ $= 18^\circ \checkmark$		1M 1A	(2)
1.3.3	Men = $\frac{21\ 000}{90\ 000} \times 360^\circ \checkmark$ $= 258^\circ \checkmark$	Men = $360^\circ - 84^\circ - 18^\circ \checkmark$ $= 258^\circ \checkmark$	1M 1A	(2)

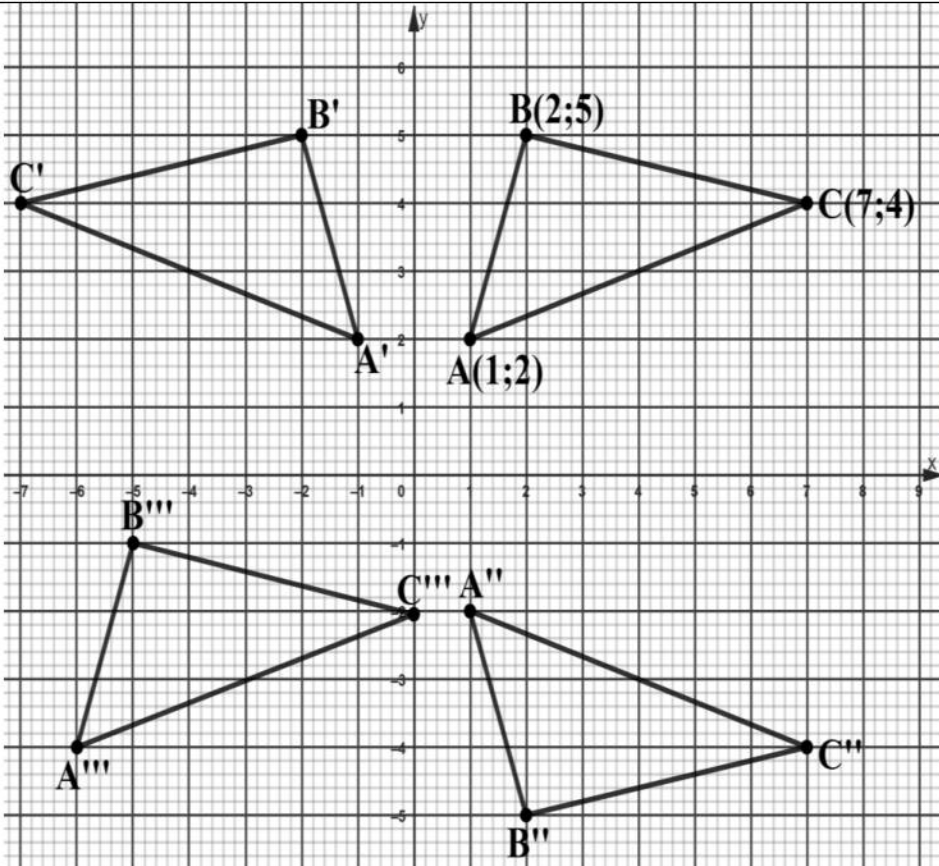
1.4				Explanation	Mark																		
	1.4.1	M = 32 ✓		1A	(1)																		
	1.4.2		<table><tr><th>STEM</th><th>LEAF</th></tr><tr><td>1</td><td>22 ✓</td></tr><tr><td>2</td><td>222✓</td></tr><tr><td>3</td><td>2222✓</td></tr><tr><td>4</td><td>222✓</td></tr><tr><td>5</td><td>22✓</td></tr><tr><td>6</td><td>222✓</td></tr><tr><td>7</td><td>22✓</td></tr><tr><td>8</td><td>2✓</td></tr></table>	STEM	LEAF	1	22 ✓	2	222✓	3	2222✓	4	222✓	5	22✓	6	222✓	7	22✓	8	2✓	1 mark for each row	(8)
STEM	LEAF																						
1	22 ✓																						
2	222✓																						
3	2222✓																						
4	222✓																						
5	22✓																						
6	222✓																						
7	22✓																						
8	2✓																						
	1.4.3	$\bar{x} = \frac{12 + 12 + \cdots + 72 + 82}{20} \quad \checkmark$ $= \frac{860}{20} \text{ OR } 43 \quad \checkmark$		1SF 1A	(2)																		

1.5			Explanation	Mark
1.5.1	A quartile is used to divide a data set into four equal groups. ✓		1A	(1)
1.5.2	A percentile is used to divide a data set into 100 equal groups. ✓		1A	(1)
				[40]

QUESTION 2

2.1		Working	Explanation	Mark
	2.1.1	$S_A = 7(3 + 4 + 5) + 2\left(\frac{1}{2} \times 4 \times 3\right) \checkmark$ $= 96 \text{ cm}^2 \checkmark$	1SF 1A	(2)
	2.1.2	$S_B = 7(3 + 4 + 5) + 2\left(\frac{1}{2} \times 4 \times 3\right) \checkmark$ $= 384 \text{ cm}^2 \checkmark$	1SF 1A	(2)
	2.1.3	$S_A : S_B = 96 : 384 \checkmark$ $\frac{S_A}{S_B} = \frac{96}{384} = \frac{1}{4} \checkmark \checkmark$ $= 1:4 \text{ OR } 1:2^2 \checkmark$	1M 1A	(2)
	2.1.4	When all dimensions of a prism are multiplied by a constant factor k, the volume is multiplied by k^2 .	1S	(1)

2.2		Working	Explanation	Mark
	2.2.1	$N = \left(\frac{-6 + 4}{2}; \frac{6 - 4}{2}\right)$ $= (-1; 1) \checkmark \checkmark$	1SF 1A	(2)
	2.2.2	$\Rightarrow -1 = \frac{y - 3}{2} \checkmark$ $\Rightarrow -2 = y - 3$ $\Rightarrow y = 1 \checkmark$	1SF 1A	(2)
	2.2.3	$\Rightarrow 1 = \frac{x + 5}{2} \checkmark$ $\Rightarrow 2 = x + 5$ $\Rightarrow x = -3 \checkmark$	1SF 1A	(2)
	2.2.4	$AB = \sqrt{(-6 - -3)^2 + (6 - -3)^2} \checkmark$ $= \sqrt{90} \checkmark$ $= 3\sqrt{10} \checkmark$	1SF 1M 1A	(3)
	2.2.5	$m_{CD} = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{1 - -4}{5 - 4} \checkmark$ $= 5 \checkmark$	1SF 1A	(2)

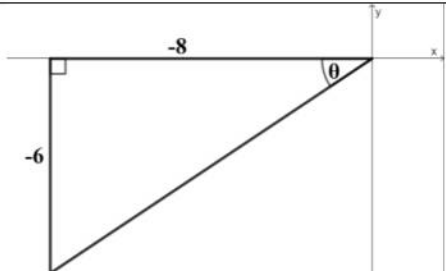
2.3	Working	Explanation	Mark
2.3.1		1 mark for each coordinate plotted correctly ✓✓✓	(3)
2.3.2	See Question 2.3.1 ✓✓✓		(3)
2.3.3	See Question 2.3.1 ✓✓✓		(3)
2.3.4	P(2;1) ✓ Q(5;2)✓ and R(4; 7)✓	1 A; 1A; 1A	(3)
			[30]

QUESTION 3

3.1	Working	Explanation	Mark
3.1.1	The period is the time a graph takes to repeat itself. ✓	1S	(1)
3.1.2	The amplitude is the maximum distance of a graph above or below the x – axis. ✓	1S	(1)

3.2	Working	Explanation	Mark
3.2.1	A = 3; B = 1; C = 0; D = 2 and E = 4 ✓	1A	
	P = 1; Q = 2; R = -1; S = -2; and T = -1 ✓	1A	(2)
3.2.2	See ADDENDUM C below ✓✓✓✓	4CA	(4)

3.2.3	$A_1 + A_2 = 4 + 2\checkmark$ $= 6\checkmark$	1M 1A	(2)
3.2.4	$x = 60^\circ\checkmark$ and $x = 300^\circ\checkmark$	1A; 1A	(2)

3.3	Working	Explanation	Mark
3.3.1		1A	(1)
3.3.2	$r = \sqrt{(-6)^2 + (-8)^2}\checkmark$ $= 10\checkmark$	1SF 1A	(2)
3.3.3	$3\sin\theta - \cos\theta = 3\left(\frac{-6}{10}\right) - \frac{-8}{10}\checkmark$ $= \frac{-18}{10} + \frac{8}{10}\checkmark$ $= -1\checkmark$	1SF 1M 1A	(3)
3.3.4	$2\sin\theta\cos\theta - \frac{2}{\tan\theta} = 2\left(\frac{-6}{10}\right)\left(\frac{-8}{10}\right) - \frac{2}{6/8}\checkmark$ $= \left(\frac{96}{100}\right) - \left(\frac{16}{6}\right)\checkmark$ $= -\frac{128}{75}$ OR $-1,707\checkmark$	1SF 1M 1A	(3)

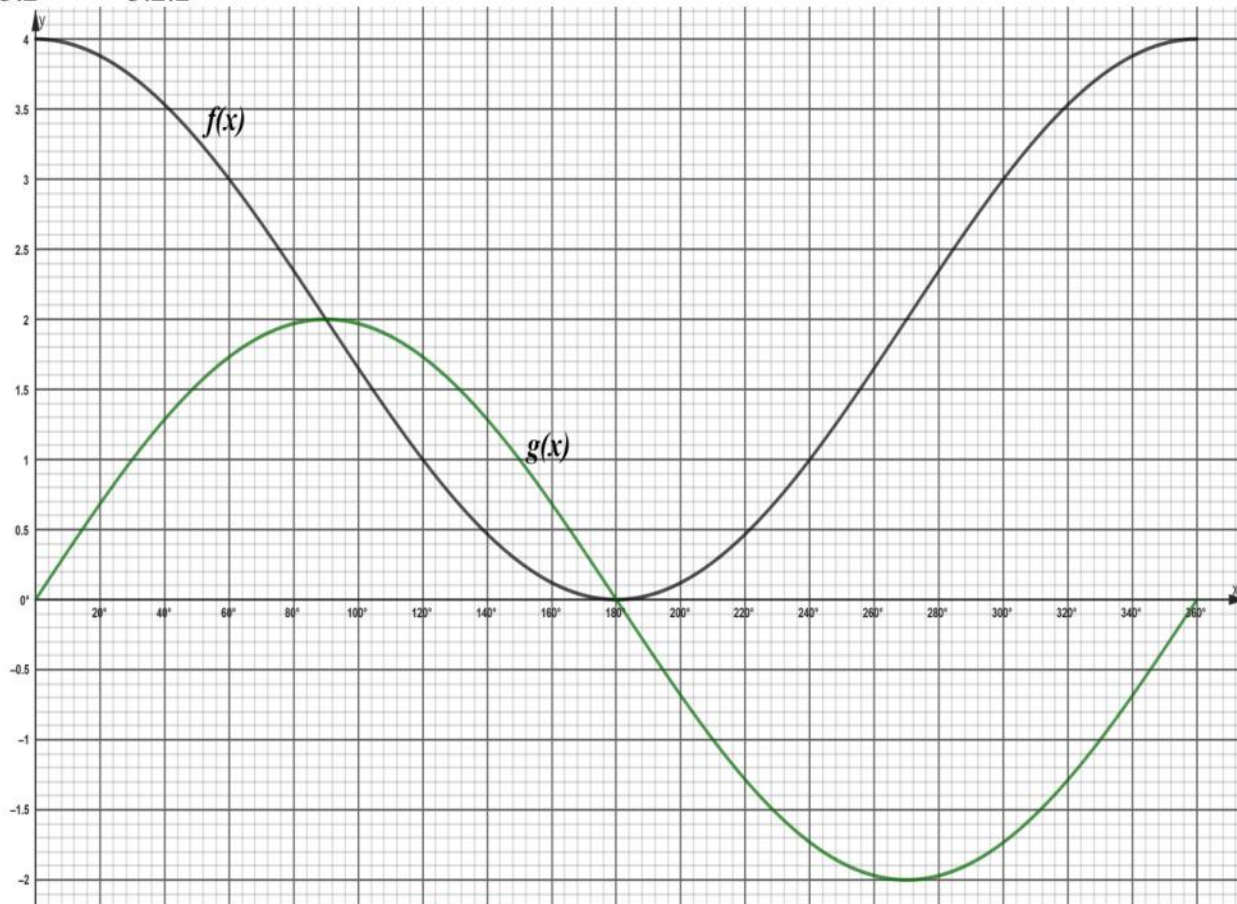
3.4	$\sin(\theta - 12^\circ) = \frac{1}{2}$ $\Rightarrow \theta - 12^\circ = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ\checkmark$ $\Rightarrow \theta = 42^\circ\checkmark$	1M 1A	(2)
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3.5		Working			Explanation	Mark
	3.5.1	$x = \sqrt{(34)^2 - (26)^2} \checkmark$ $= 4\sqrt{30} \text{ cm OR } 21,909\text{cm} \checkmark$			1SF 1A	(2)
	3.5.2	$y = \sqrt{(4\sqrt{30})^2 - (20)^2} \checkmark$ $= 4\sqrt{5} \text{ cm OR } 8,944\text{cm} \checkmark$			1SF 1A	(2)
	3.5.3	$\tan \alpha = \frac{20}{4\sqrt{5}} \checkmark$ $\alpha = \tan^{-1}\left(\frac{20}{4\sqrt{5}}\right)$ $= 65,905^\circ \checkmark$	$\sin \alpha = \frac{20}{4\sqrt{30}} \checkmark$ $\alpha = \sin^{-1}\left(\frac{20}{4\sqrt{30}}\right) \checkmark$ $= 65,905^\circ \checkmark$	$\cos \alpha = \frac{4\sqrt{5}}{4\sqrt{30}} \checkmark$ $\alpha = \cos^{-1}\left(\frac{\sqrt{5}}{\sqrt{30}}\right) \checkmark$ $= 65,905^\circ \checkmark$	1SF 1M 1A	(3)
						[30]

ADDENDUM C

3.2

3.2.2



Shape of $f(x)$ ✓

Shape of $g(x)$ ✓

$(180^\circ, 0)$ for $f(x)$ ✓

$(90^\circ, 2); (180^\circ, 0)$ and $(270^\circ, -2)$ for $g(x)$ ✓

TOTAL: 100